

Project: E-Commerce Sales Dashboard

- **Objective:**

This project served as an introductory exercise to Power BI, focusing on fundamental data import, transformation, and visualization techniques. The primary objective was to understand the end-to-end process of creating an interactive dashboard from raw data to actionable insights, using publicly available e-commerce sales data.

- **Problem Statement:**

As a beginner, the core problem was to grasp the capabilities of Power BI in transforming raw transactional data into meaningful business insights. This involved learning how to connect to data sources, clean and model data, and design effective visualizations to answer basic business questions related to sales performance, profit trends, and customer behaviour.

- **Data Sources Used:**

The dashboard utilized two primary CSV files:

- **Details.csv:** Contained transactional details such as product categories, sub-categories, quantities, and amounts.
- **Orders.csv:** Contained order-level information including dates, payment modes, and geographical data (State, City).

These files were sourced from the internet for learning purposes.

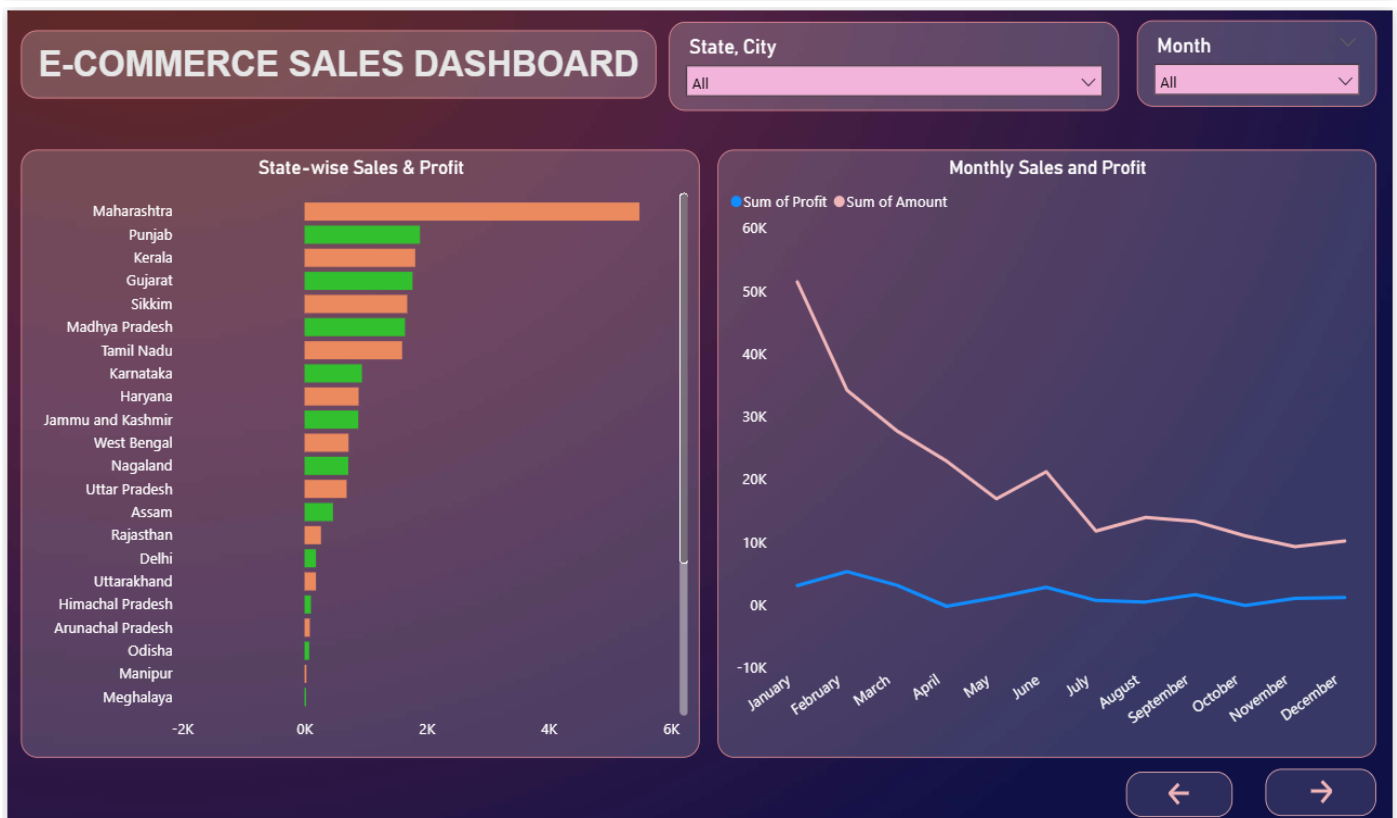
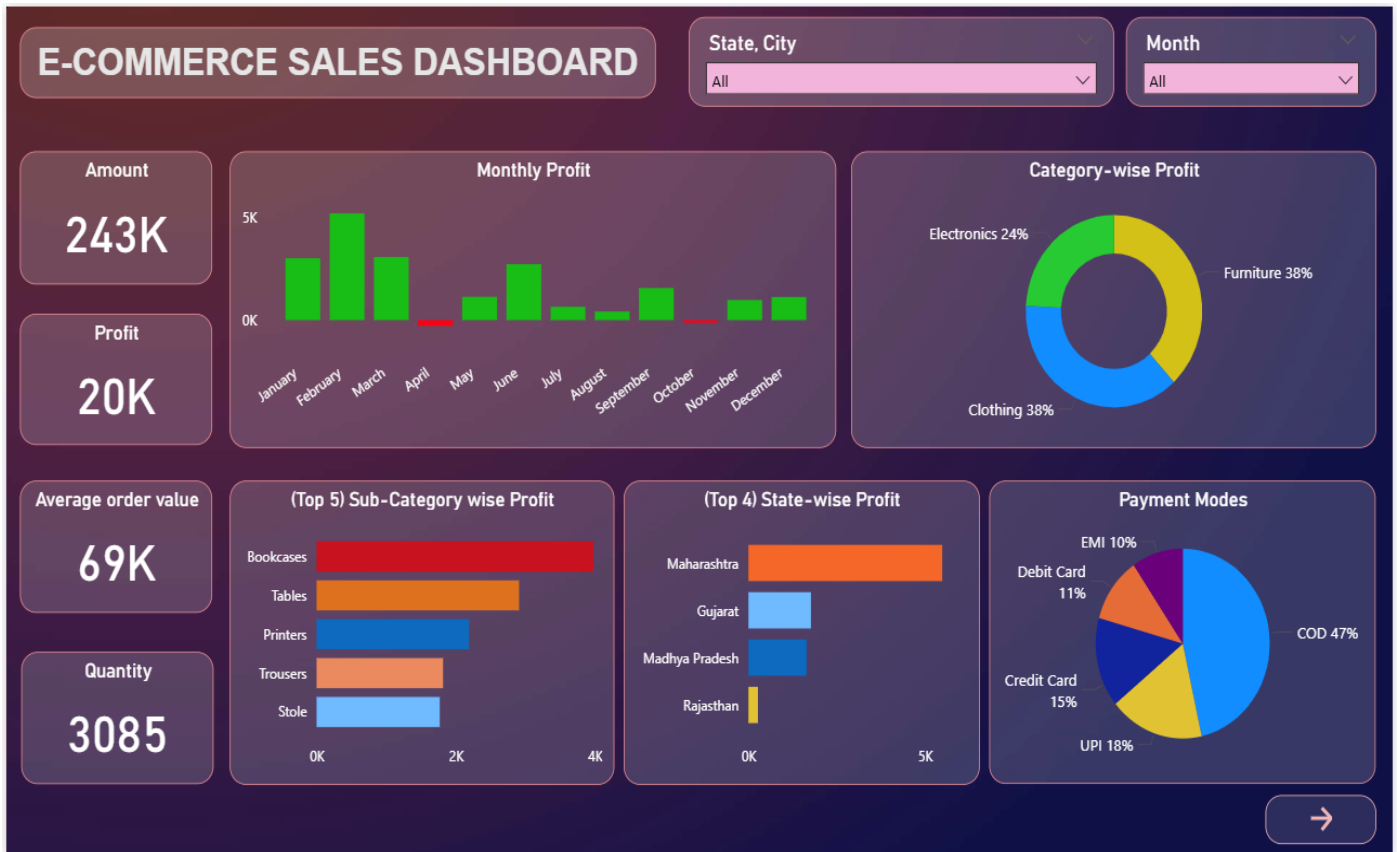
- **Key Metrics and KPIs Addressed:**

The dashboard effectively visualized several key performance indicators (KPIs) and metrics, including:

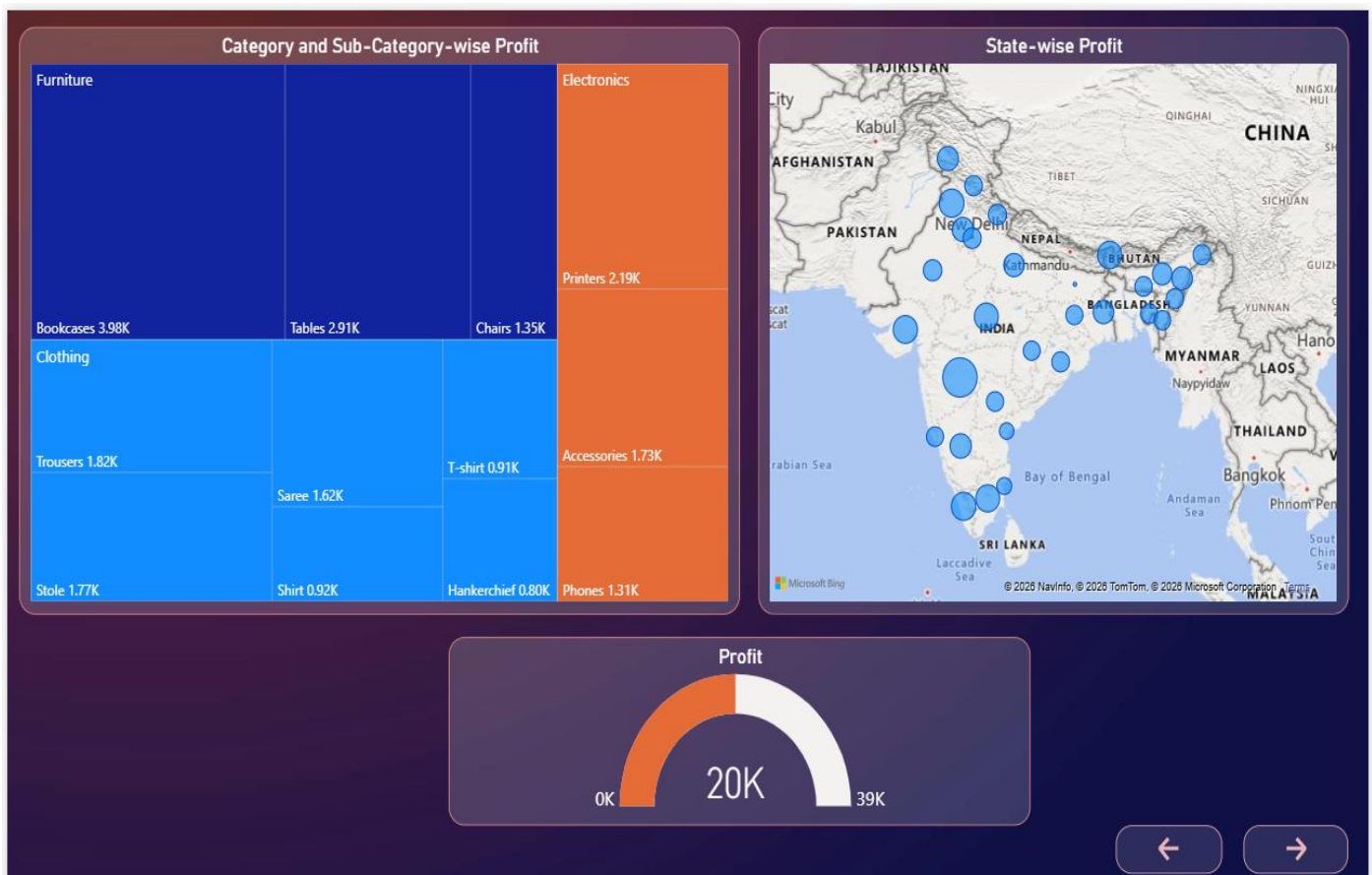
- **Total Amount & Profit:** Overall sales volume and profitability.
- **Average Order Value:** Insight into the typical value of each transaction.
- **Quantity:** Total number of items sold.
- **Monthly Profit:** Trend analysis of profit over different months.
- **Category-wise Profit:** Breakdown of profit by product categories (e.g., Electronics, Furniture, Clothing).
- **Sub-Category-wise Profit:** More granular view of profit within sub-categories (e.g., Bookcases, Tables, Printers).
- **State-wise Sales & Profit:** Geographical performance analysis.
- **Payment Modes:** Distribution of sales across different payment methods (e.g., COD, Debit Card, Credit Card, UPI).

- **Monthly Sales and Profit Trends:** Line charts showing the progression of sales and profit throughout the year.
- **Category-wise Quantity and Profit:** Comparative analysis of profit and quantity across different product categories.
- **Sales Trend (Daily/Monthly):** Detailed daily and monthly sales patterns.
- **Financial Year Sales Trend:** Comparison of sales performance across different financial years.

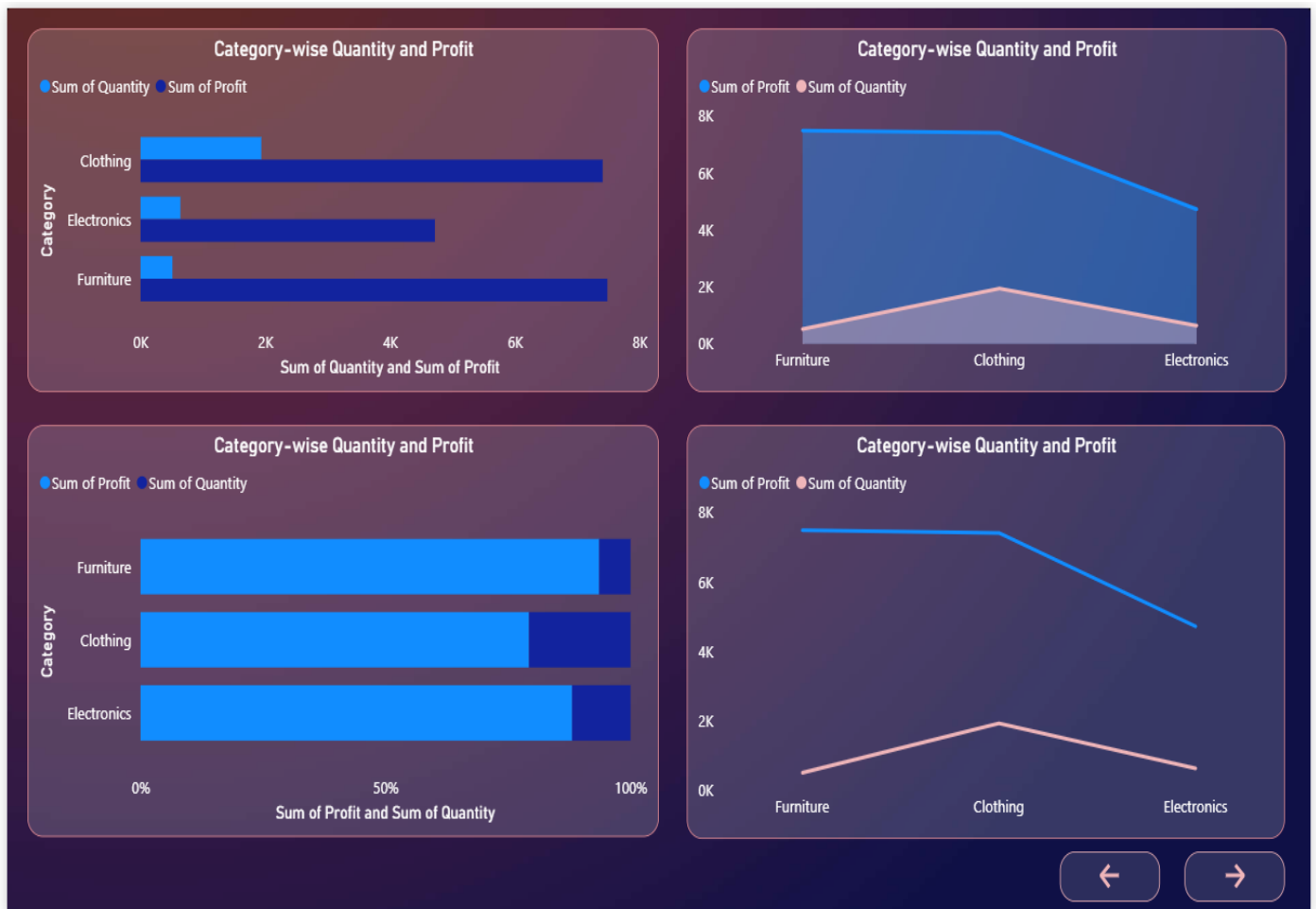
- Screenshots:



The initial dashboards provide an executive summary of sales performance, showing total amount, profit, average order value, and quantity. They also include monthly profit trends, category-wise and sub-category-wise profit distribution, and payment mode analysis, along with state-wise sales and profit, and comprehensive monthly sales and profit trend line charts. Further detailed views illustrate category-wise quantity and profit comparisons using various chart types, including stacked bar charts and area charts, showcasing different ways to visualize the same data for varied insights.



The waterfall chart highlights the contribution of each category to the total profit, while the streamgraph visualizes profit distribution across categories and payment modes. Geographical insights are further expanded with a map visualization showing state-wise profit, complemented by a treemap displaying category and sub-category profit.



Category-wise Profit

Clothing

Electronics

Furniture

Sum of Amount	State	Sum of Profit
6854	Andhra Pradesh	-494
497	Arunachal Pradesh	86
2577	Assam	465
6883	Bihar	-1357
2687	Chhattisgarh	-53
9766	Delhi	187
8442	Goa	-50
11446	Gujarat	1768
6975	Haryana	885
3394	Himachal Pradesh	105
11001	Jammu and Kashmir	880
2299	Jharkhand	-38
7019	Karnataka	940
9436	Kerala	1812
34476	Madhya Pradesh	1645
243433		19605

State and City-wise Profit

City Abohar Adilabad Adityapur Agartala



Category



Year

2018

2019

Month

January

February

March

April

May

June

July

August

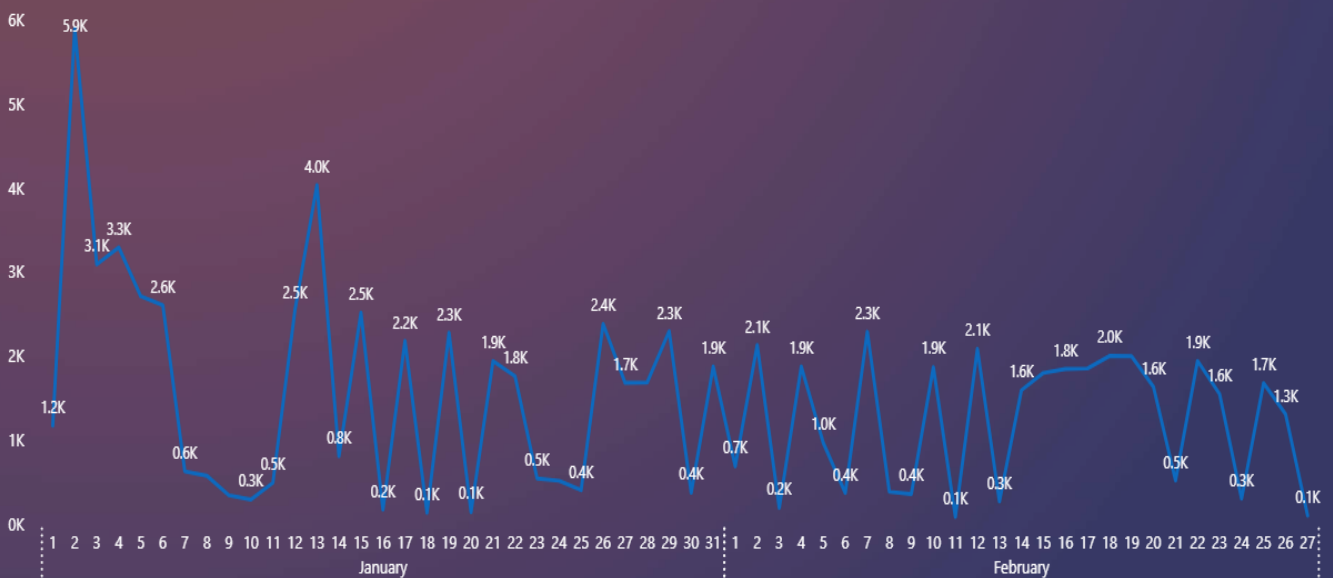
September

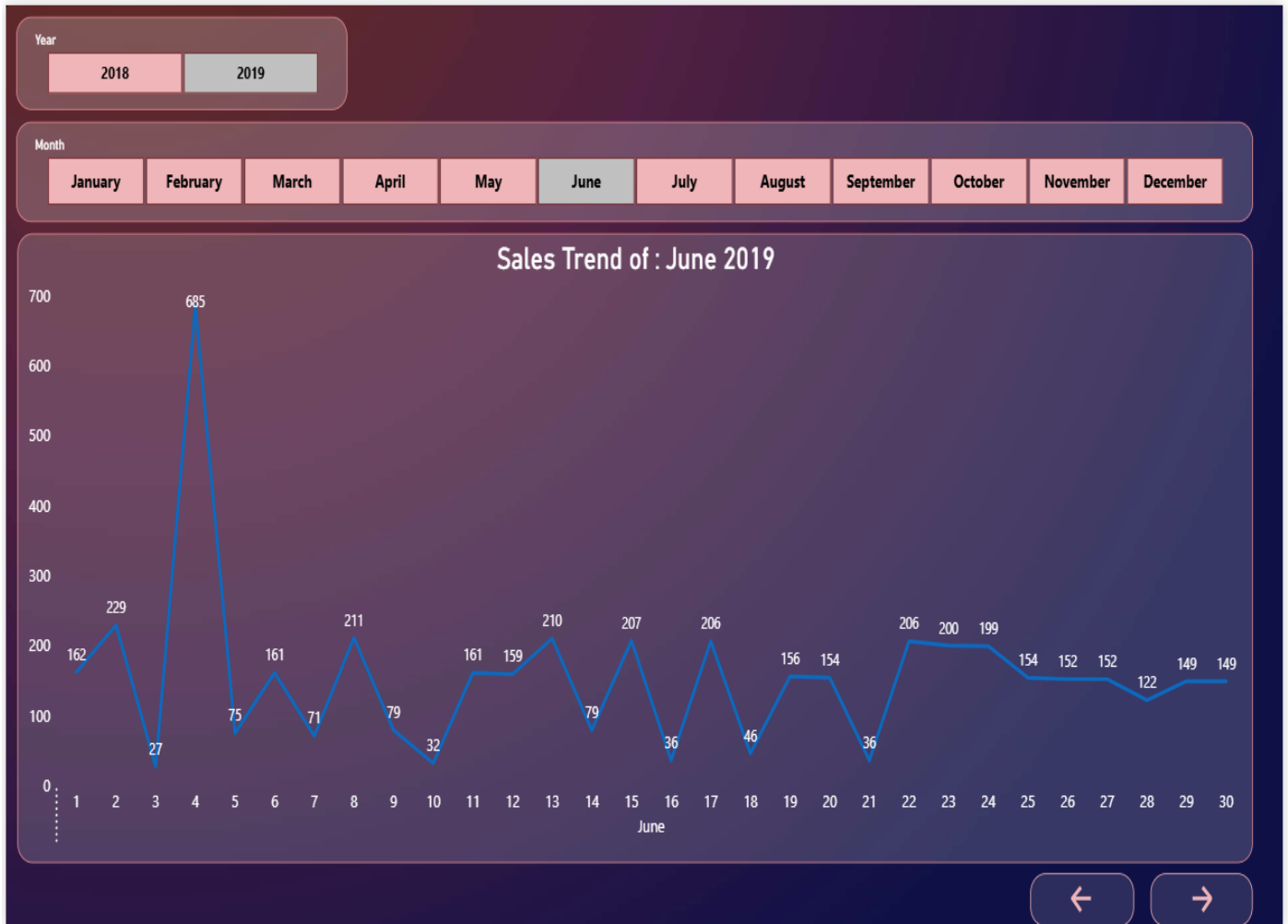
October

November

December

Sales Trend of :





Sum of Profit	Sum of Amount	Sum of Quantity
20K	243K	3K

Category-wise Profit and Quantity	
7481✓	
Goal: 511 (+1,363.99%)	

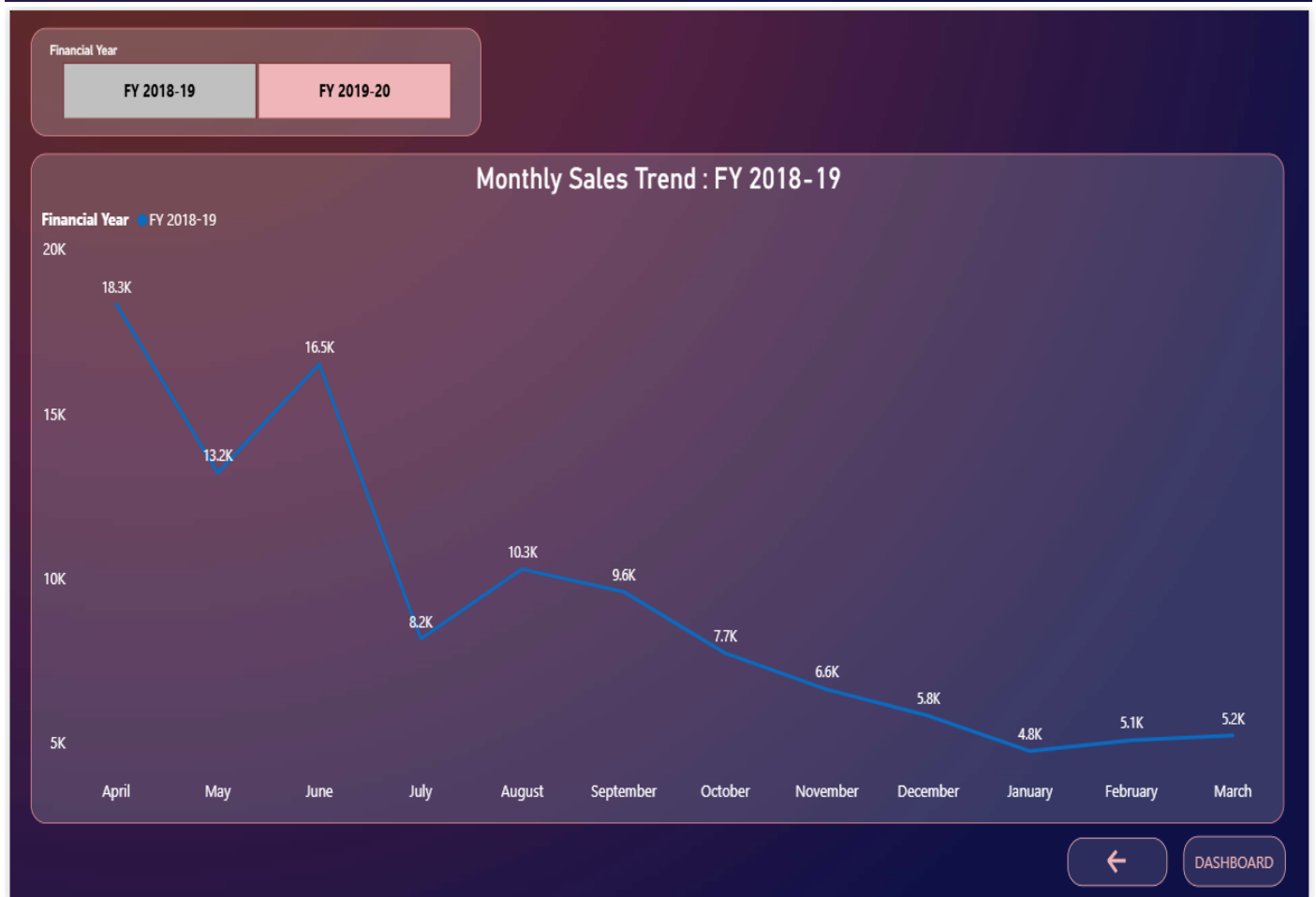
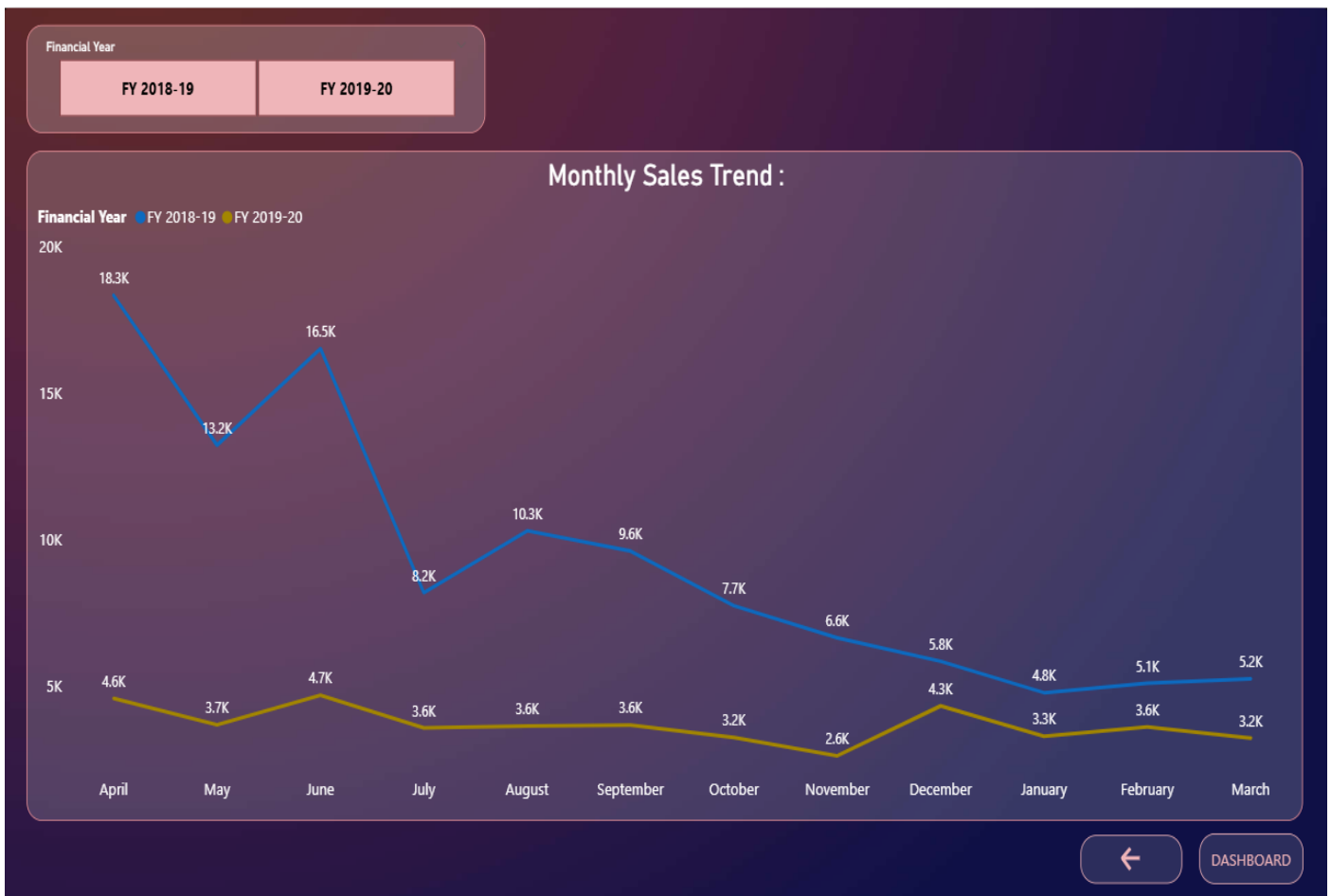
Clothing	7406	1935
Sum of Profit		Sum of Quantity
Electronics	4718	639
Sum of Profit		Sum of Quantity
Furniture	7481	511
Sum of Profit		Sum of Quantity

Sub-Category	Accessories		Bookcases		Chairs	
Month	Sum of Amount	Sum of Quantity	Sum of Amount	Sum of Quantity	Sum of Amount	Sum of Quantity
January	439	10	8028	24	8571	
February	3154	37	223	2	2392	
March	1301	5	3451	14	435	
April	1909	16	3947	26	2084	
May	316	10	1045	4	2523	
June			4883	23	516	
July	624	11			949	
August	326	7	1541	6	942	
September	737	14	2115	17	210	
October	389	7	1177	12	830	
November	42	1	597	4	890	
December	253	4	865	7	1644	
Total	9490	122	27872	139	21986	

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Detailed tabular data for sub-category performance across months offers granular insights into monthly sales and quantity for specific product lines. The daily sales trend provides a granular view of sales fluctuations within months, while other chart drills down into city-wise profit within a selected state, highlighting profitable and unprofitable regions.



Finally, the monthly sales trend across financial years allows for year-over-year performance comparison, demonstrating the ability to analyse historical data.

- **Explanation of the Visualizations and their Insights:**

- **Card Visuals:** Clearly display high-level KPIs like total amount, profit, average order value, and quantity, providing an immediate snapshot of performance.
- **Bar Charts (Monthly Profit, State-wise, Sub-Category-wise):** Effectively illustrate performance across different dimensions, highlighting top-performing months, states, and product sub-categories. Negative profit values are clearly visible, indicating areas for improvement.
- **Donut/Pie Charts (Category-wise Profit, Payment Modes):** Show the proportional distribution of profit across categories and the popularity of various payment methods, aiding in strategic decision-making.
- **Line Charts (Monthly Sales and Profit, Daily Sales Trend, Monthly Sales Trend by Financial Year):** Provide insights into trends and seasonality, allowing for identification of peak periods, dips in performance, and year-over-year comparisons. The daily trend helps in understanding short-term fluctuations.
- **Stacked Bar Charts/Area Charts (Category-wise Quantity and Profit):** Offer comparative views of multiple metrics across categories, showing both absolute values and their proportions.
- **Waterfall Chart (Category-wise Profit):** Visually represents the positive and negative contributions of each category to the total profit, making it easy to see the net effect.
- **Streamgraph (Category-wise Profit and Payment Modes):** Illustrates the flow and distribution of profit across categories, broken down by payment methods, providing a rich, multi-dimensional view.
- **Treemap (Category and Sub-Category-wise Profit):** Displays hierarchical data, where the size of the rectangle corresponds to the profit, allowing for quick identification of the most profitable categories and sub-categories.
- **Map Visual (State-wise Profit):** Provides a geographical representation of profit, helping to identify strong and weak performing regions at a glance.
- **Table Visuals:** Offer detailed, granular data for specific dimensions (e.g., State and City-wise Profit, Sub-Category monthly data), allowing users to examine exact figures.
- **Slicers (State, City, Month, Quarter, Year, Financial Year):** Enable interactive filtering, allowing users to drill down into specific data points and gain more granular insights, enhancing the dashboard's interactivity and usability.

- **Challenges Faced and How I Overcame Them:**

- **Data Cleaning and Transformation:** Initial challenge involved cleaning raw CSV data, handling missing values, and ensuring correct data types using Power Query. This was overcome by leveraging Power Query Editor's functionalities for data shaping and transformation.
- **Data Modelling:** Understanding relationships between Details.csv and Orders.csv was crucial. This was addressed by creating appropriate relationships in the Power BI data model to ensure accurate calculations across tables and enable cross-filtering.

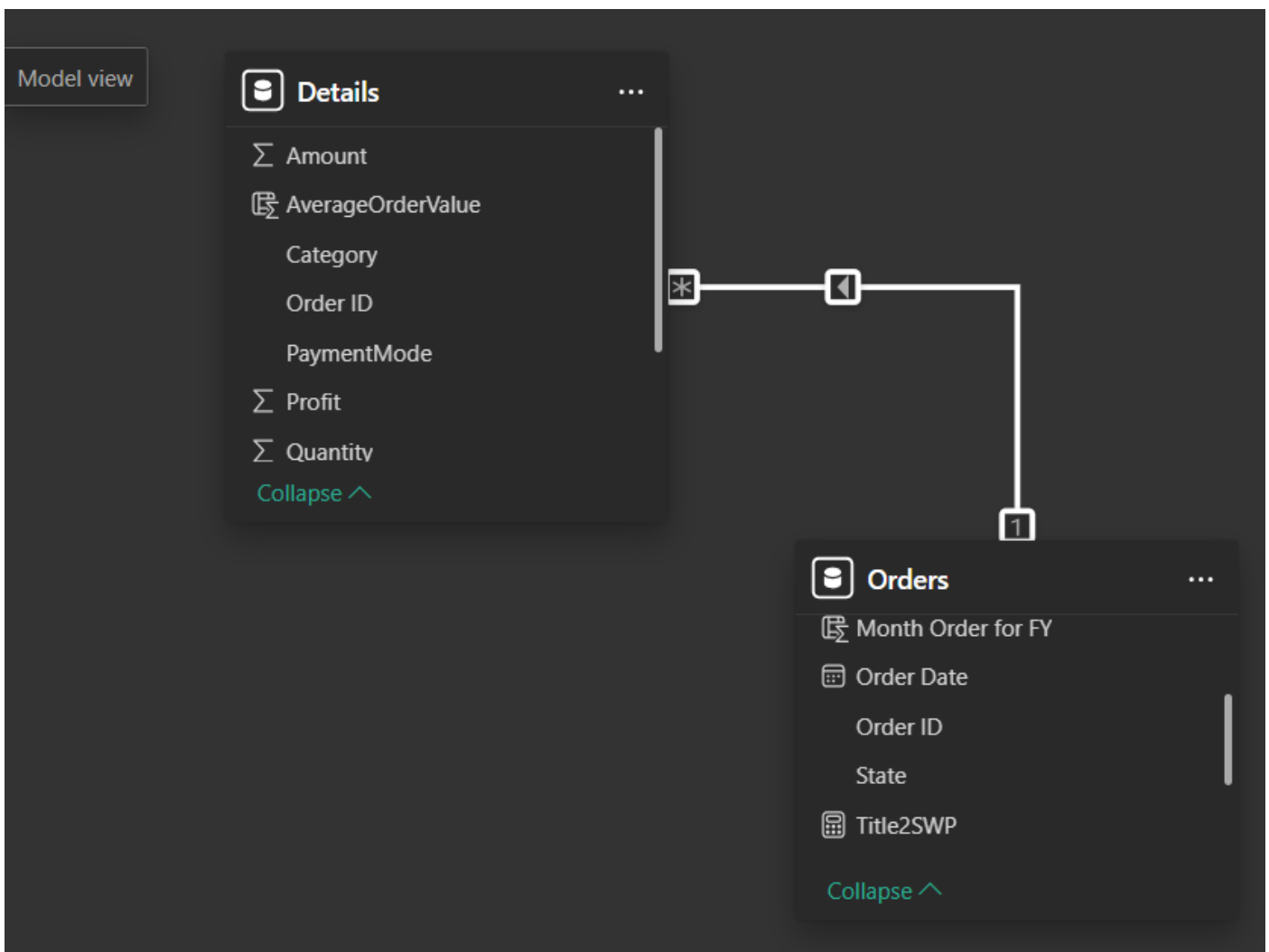
- **DAX Formulas:** Learning basic and intermediate DAX (Data Analysis Expressions) for creating measures like "Profit," "Average Order Value," and various sum/quantity calculations was a challenge. This was overcome through practice, referring to Power BI documentation, and utilizing online resources and community forums.
- **Visualization Selection and Design:** Selecting appropriate visualization types for different data stories and arranging them for a clear, intuitive, and aesthetically pleasing dashboard layout was a learning curve. This was improved through experimentation, applying best practices for dashboard design, and iterative refinement based on visual feedback.
- **Interactivity and Navigation:** Implementing interactive elements like slicers and navigation buttons between different dashboard views required understanding Power BI's action settings. This was achieved by configuring button actions and filter interactions effectively.
- **Impact/Benefits of the Dashboard:**

This learning project provided a strong foundational understanding of Power BI. It demonstrated the ability to:

 - Connect and transform diverse data sources efficiently.
 - Create a robust and well-structured data model.
 - Develop multiple interactive and insightful dashboards catering to various analytical needs.
 - Extract key business metrics, identify trends, and pinpoint areas of strength and weakness (e.g., unprofitable states/cities, underperforming categories).
 - Lay the groundwork for more complex business intelligence projects and real-world data analysis challenges. The primary benefit was the comprehensive acquisition of essential Power BI skills, crucial for future data analysis tasks and contributing to data-driven decision-making.

TOOLS & TECHNOLOGIES USED

- During this internship, the primary tool utilized was Microsoft Power BI Desktop. My work involved extensive use of its core functionalities:
- **Power Query Editor:** For data extraction, transformation, and loading (ETL) from various sources (e.g., Excel, CSV files). This included data cleaning, shaping, merging, and appending queries to prepare data for analysis. (You can embed screenshots here if you have any relevant Power Query examples. Since you didn't provide any for Power Query specifically, you can either omit this screenshot note or add one if you find a suitable image.)
- **Data Modelling:** Designing and implementing robust data models, establishing relationships between tables, and optimizing data for performance and analytical flexibility.
- **Explanation:** As a foundational step, I established a 'one-to-many' relationship between the Details table (containing transactional details) and the Orders table (containing order-level information). This relationship, based on a common Order ID, was crucial for enabling accurate cross-filtering and calculations across the dataset.
- **Screenshot:**



- **DAX (Data Analysis Expressions):** I extensively used DAX for creating complex measures and calculated columns to derive key performance indicators (KPIs) and enable in-depth analysis. This included both aggregation measures and dynamic text measures for enhanced report interactivity.
- **Explanation (DAX Formulas used in Project: E-COMMERCE SALES DASHBOARD):** I developed various DAX measures to aggregate data (e.g., Total Sales) and to create dynamic titles that respond to user selections (e.g., Title2SWP, Title3FY, TitleMP, TitleMY). These dynamic titles improve the user experience by providing context-sensitive information.

Project DAX Formulas:

1. Title2SWP (Dynamic State-wise Profit Title):

```
1 Title2SWP = "State-wise Profit : " & SELECTEDVALUE(Orders[State])
```

Purpose: Dynamically displays the selected state in the title of a "State-wise Profit" visualization. If no state is selected, it will likely show a blank or the default value of SELECTEDVALUE.

2. Title3FY (Dynamic Monthly Sales Trend Title by Financial Year):

```
1 Title3FY = "Monthly Sales Trend : " & SELECTEDVALUE(Orders[Financial Year])
```

Purpose: Dynamically displays the selected financial year in the title of a "Monthly Sales Trend" visualization.

3. TitleMP (Dynamic Monthly Profit Title):

```
1 TitleMP = "Monthly Profit " & SELECTEDVALUE(Orders[Order Date].[Month])
```

Purpose: Dynamically displays the selected month in the title of a "Monthly Profit" visualization.

4. TitleMY (Dynamic Sales Trend Title by Month and Year):

```
1 TitleMY = "Sales Trend of : " & SELECTEDVALUE(Orders[Order Date].[Month]) &  
" " & SELECTEDVALUE(Orders[Order Date].[Year])
```

Purpose: Dynamically displays the selected month and year in the title of a "Sales Trend" visualization, providing context for the displayed data.

5. Total Sales (Aggregation Measure):

```
1 Total Sales = SUM(Details[Amount])
```

Purpose: Calculates the total sales amount by summing the Amount column from the Details table. This is a primary revenue metric.

CONCLUSION

The E-Commerce Sales Dashboard project successfully demonstrated the complete workflow of developing an interactive business intelligence solution using Microsoft Power BI. Through this project, raw transactional and order-level data were transformed into meaningful insights using data cleaning, modelling, DAX calculations, and effective visualizations. The dashboard provided a comprehensive view of key business metrics such as sales, profit, quantity sold, average order value, payment mode distribution, and geographical performance.

By incorporating interactive filters, dynamic titles, and a variety of visualizations, the dashboard enabled users to analyse trends, compare performance across categories and regions, and identify profitable as well as underperforming areas. The project also strengthened practical skills in Power Query, data modelling, DAX, and dashboard design while emphasizing the importance of data-driven decision-making.

Overall, this project served as a strong foundation in Power BI and business analytics, enhancing my ability to transform raw data into actionable insights and preparing me for more advanced data visualization and reporting projects in the future.